

Python Programming Assignment

1. Write a program that accepts a list of integers and uses Pool.map() to calculate the sum of squares from 1 to N for every element in the list.

Example Input

```
[1000000,2000000,3000000,4000000]
```

Expected Output

```
[333333833333500000,  
2666668666667000000,  
...  
]
```

2. Write a program that calculates factorials of multiple numbers simultaneously using Pool.map().

Input

```
[10,15,20,25]
```

Display

- Process ID
- Input Number
- Factorial

3. For every number in the given list, count how many prime numbers exist between 1 and N using multiprocessing Pool.

Example

```
10000  
20000  
30000  
40000
```

Display total prime count for each number.

4. Write a program that calculates

$$1^5+2^5+3^5+\dots+N^5$$

for multiple values of N simultaneously using Pool.

Input

```
[1000000,  
2000000,  
3000000,  
4000000]
```

Measure total execution time.

